Project: Light & Motion Security Module

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1. Project Purpose

This project's goal is to create a simple, effective, and low-cost security module using an ESP32 microcontroller. The module is designed to detect two key environmental changes: the presence of motion and a drop in ambient light levels. It uses visual indicators (LEDs) to provide immediate alerts without the need for a separate display, making it suitable for a wide range of applications such as home security, automated lighting, or simple event monitoring in a small area.

2. Sensors and Indicators

The module integrates the following components to achieve its functionality:

- **PIR (Passive Infrared) Motion Sensor:** This sensor detects changes in infrared radiation, which is typically emitted by warm bodies. When motion is detected, its digital output goes HIGH.
- **LDR (Light-Dependent Resistor):** This sensor measures ambient light levels. Its resistance changes based on the amount of light it receives. The ESP32 reads the LDR's analog value, which can then be converted to a digital signal (e.g., if the value is below a certain threshold, it is considered "low light").
- **Red LED:** This acts as a visual alert for motion detection. It illuminates when the PIR sensor detects movement.
- **Yellow LED:** This indicates low light conditions. It turns on when the LDR sensor detects that ambient light has dropped below a set threshold.

3. ESP32 Pin Assignment

The following pins on the ESP32 are used to interface with the module's PCB and components.

These are chosen for their versatility and common usage on the ESP32 development board.

Component	ESP32 Pin	Type	Notes
PIR Sensor	GPIO 13	Digital Input	A logic HIGH signal indicates motion is detected.

Red LED	GPIO 2	Digital Output	`HIGH` to turn on the LED, `LOW` to turn it off.
LDR Sensor	GPIO 34	Analog Input	ESP32's ADC pin for reading variable resistance.
Yellow LED	GPIO 4	Digital Output	`HIGH` to turn on the LED, `LOW` to turn it off.

The sensors and LEDs are connected to the ESP32's VCC (3.3V) and GND pins to provide power.

4. System Logic and Operation

The microcontroller's program continuously monitors the state of both sensors. The core logic operates as follows:

- **Motion Detection:** The program reads the digital state of the PIR sensor on GPIO 13. If the state is HIGH, it immediately sets the red LED (GPIO 2) to HIGH, turning it on. The LED remains on for a short period before being turned off, or as long as motion is detected.
- **Low Light Detection:** The program reads the analog value from the LDR on GPIO 34. This value is compared against a pre-defined threshold. If the reading is below this threshold, it indicates low light, and the program sets the yellow LED (GPIO 4) to HIGH, turning it on.

The two alert functions operate independently, meaning that both LEDs can be on at the same time if both motion and low light are detected.

5. Screenshots

Here are some conceptual screenshots to visualize the project's development and final assembly.

PCB Design and Layout: This shows the final PCB design before fabrication, with all component footprints and traces laid out.

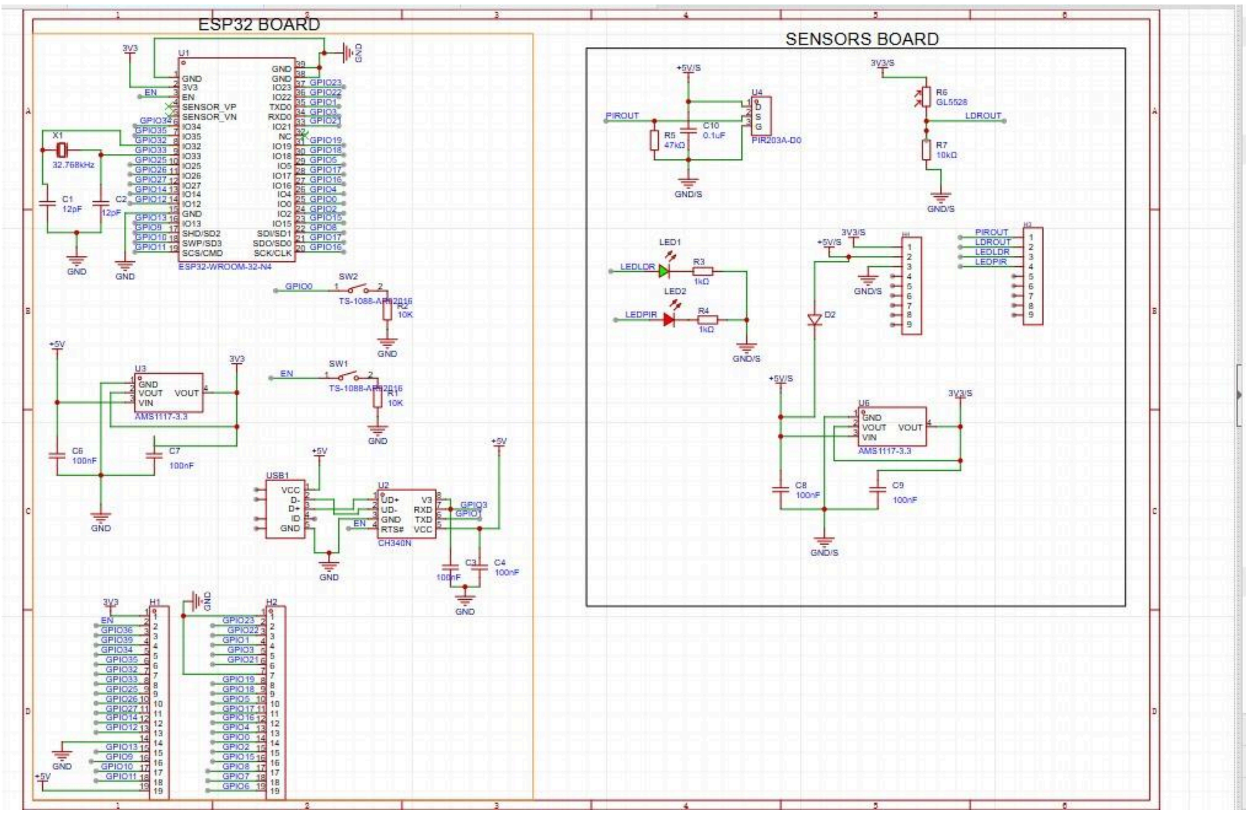
Final Assembled Module: This demonstrates the completed hardware setup, with the sensors and LEDs soldered to the custom PCB and connected to the ESP32.

Serial Monitor Output: This shows the debugging output from the ESP32, displaying the current status of the sensors (e.g., "Motion Detected!" or "Low Light Detected").

Documents

Board1

Schematic



P1

PCB